UDACITY

Cybersecurity Foundations



Securing a Computer System

Securing a Computer System



Congratulations!

You have been hired to audit the security for the PC used at your friend's business: Joe's Auto Body. Joe provides car repair services throughout the tri-state area. He's had previous employees use it for activities unrelated to work (e.g., web browsing, personal email, social media, games, etc.), and he now uses it to store his critical business information. He suspects that others may have broken into it and could be using it to transfer files across the internet. He has asked that you secure it for him according to industry best practices so that it can again be used as a standard PC.

In this project, you have been given a "broken" Windows 10 PC and asked to figure out what's wrong with it and then make changes to fix and secure it. The process of analyzing and applying security happens in workplaces around the globe and is exactly what cybersecurity professionals do daily. This project allows you to apply what you've learned in the course by investigating a Windows 10 PC. The same skills you use on one PC can be applied to thousands.

You do not need to do anything on this slide.

Part 1: Reconnaissance

Hardware

The first step in securing any system is to know what it is, what's on it, what it's used for, and who uses it. That's the concept of systems reconnaissance and asset inventory. In this step, you'll document the hardware, software, user access, system and security services on the PC. Complete each section below.

1	*Device Name*	JoesGaragePC
2	*Processor*	Intel® Xeon® CPU E5-2673 v4 @ 2.30 GHz 2.29 GHz
3	*Install RAM*	4.00 GB
4	*System Type*	64-bit operating system, x64-based processor
5	*Windows Edition*	Windows 10 Pro
6	*Version*	22H2
7	*Installed on*	11/23/2021
8	*OS build*	19045.2486

Software

Another common early step in securing is taking an inventory of software or applications installed on a computer system. These are programs outside of the standard operating system. Please list five applications running on this PC.

1	Farm Heroes Saga
2	Candy Crush Friends
3	Hotspot Shield
4	Streaming Audo Recorder Plus
5	MusicBee

Accounts

As part of your security assessment, you should know the user accounts that may access the PC. Please list the accounts, name, and access level for the accounts on this PC.

Account Name	*Full Name*	*Access Level*
Frank	Frank	User with Remote Desktop Access
Hacker	A Hacker	Administrator
JaneS	Jane Smith	Administrator
JoesAuto	Joes Account	Administrator
Notadmin	Do Not Use	User with Remote Desktop Access
AUser	A User	User Access

Security Services

Document the PC's security settings status listed below.

	Security Feature	*Status*
	Firewall product and status--Private network	Off
	Firewall product and status--Public network	On
	Virus protection product and status	Windows Defender, On
	Internet Security messages	On
	Network firewall messages	On
	Virus protection messages	On
	User Account Control Setting	Never notify

Part 2: Assessment

Authentication

Consider the 3 factors of authentication: something you KNOW, something you HAVE, and something you ARE. In 1 to 3 sentences below, suggest and explain what type of authentication would be appropriate for JoesPC.

1. Joe should use a strong password.
2. Joe should adopt MFA (Multi-factor Authentication), such as SMS.
3. Joe should get a fingerprint sensor or a Windows certified webcam to unlock.

System and Security: Firewall

Please answer the following question.	
1	*In 1 to 2 sentences, explain what protection would enabling the Windows Firewall provide.* Enabling the firewall would check incoming/outgoing requests to/from Joe's PC and protect it using pre-set rules.

System and Security: Firewall

Scenario: You need to ensure the Windows Defender anti-virus is enabled to always protect against current threats. It should be set to continually scan the PC for malicious software automatically. Please answer the following question.	
1	*In 1 to 2 sentences, explain what protection enabling the Microsoft Defender Antivirus would provide.* Windows Defender scans everything for viruses, malware, etc. It will also detect an ongoing threat and stop it.

Part 3: Securing Access

Users - Part 1

Ensuring only specific people have access to a computer is a common step in information security. It starts by understanding who should have access and the rules or policies that should be followed. Please review the following users who should have access.

- JoesAuto
- Jane Smith (Joe's Assistant)

It is your responsibility to create suggestions for securing this computer. Use the next slide to give and explain your recommendations. The slide following your recommendations will have two questions regarding users and privileges.

Users - Part 2

Fill in this table based on the guidelines you would recommend to Joe. Recommendations do not have to be in complete sentences. Explanations must be at least one sentence.

	Recommendation	*Explanation*
How should users authenticate their identity?	Two	Joe has to be allowed access, Jane is also his assistant so she needs access as well.
What Access Rights/Permissions should Joe have?	Administrator	He owns the computer and he should be able to make the changes.
What Access Rights/Permissions should Jane have?	User	She accesses the computer to help Joe for any reason but doesn't need any administrator privileges.

Users- Part 3

Please answer the following two questions.	
1	*In 1 to 2 sentences, explain why it is important to disable or remove unneeded accounts from a PC or application.* To prevent previous users from accessing their accounts again and revoke privileges.
2	*Administrator privileges for too many users is a security challenge. Provide at least 3 risks associated with users having administrator rights on a PC.* *1.* User (intentionally or unintentionally) does an action which exposes the computer to threats using administrator privileges. *2.* It causes the possibility for non-repudiation. Too many administrators means assigning blame is very difficult. *3.* It provides hackers with multiple accounts to try to break in to, whichever has the weakest password.

Part 4: Securing Applications

Unnecessary Applications

Joe wants everyone to use the latest version of the Internet Explorer browser by default. There should be no games or non-work-related applications installed or downloaded. Joe is also concerned that there are “hacking” programs downloaded or installed on the PC that should be removed. This PC is used for standard office functions.

1	*List three applications that violate this policy.* 1. Candy Crush Friends 2. Farm Heroes Saga 3. MusicBee
2	*Name three vulnerabilities, threats, or risks to having unnecessary applications.* 1. Possible risk of an application being hacked (generally) and working as a backdoor to the host’s PC. 2. Possibility of using a malicious application which could execute code on the host’s PC. 3. Having too many applications makes management more complex, and complexity is the enemy of security.

Patching and Updates

All applications should be up-to-date on patches or fixes by the manufacturer. Any old version of software should be uninstalled. List two applications on JoesPC that are out of date.	
1	Adobe Reader XI Version: 11.0.01 Release Date: 11/5/2020
2	7-Zip Version: 19.00 Release Date: 23/11/2021

Standout Suggestions

Standout Suggestion 1

Joe has decided to allow least privilege access to 2 additional employees. He would like the bookkeeper and the head mechanic to have access to JoesPC. In 3 - 5 sentences total below, describe the privileges these two employees should have, and detail how they should authenticate their identities.	
1	Both of them should have access to the shared work files and work applications, only user privileges, never administrator. They should use strong unique passwords, MFA (SMS/Authenticator App) and fingerprint/facial recognition.

Standout Suggestion 2

Joe believes one of his employee's emails has been compromised. What are the possible threats, risks, or vulnerabilities, and how should he respond? Detail your answer in 3 to 5 sentences.	
2	The compromised email could send malicious emails to everyone the employee knows and has contact with, causing all of them to be infected. Joe should revoke any privileges granted to the employee (temporarily) and warn everyone in work not to respond or even click on his emails, Joe should tell the employee to contact support and get his email back.